

Analysis of Variance (ANOVA)

Comparing Two Population Variances:

Assumptions of the F-Test

1. **Independence**

The two samples must be drawn independently from their respective populations.

2. **Normality**

Each population should follow a **normal distribution**.

3. **Random Sampling**

Observations in each sample are obtained through random sampling.

4. **Scale of Measurement**

The data must be **quantitative (interval or ratio scale)**.

5. **Homogeneity under H_0**

Under the null hypothesis, the population variances are assumed to be equal:

$$H_0: \sigma_1^2 = \sigma_2^2$$

Hypothesis

$$(i) H_0: \sigma_1^2 = \sigma_2^2 \text{ vs. } H_1: \sigma_1^2 > \sigma_2^2$$

$$(ii) H_0: \sigma_1^2 = \sigma_2^2 \text{ vs. } H_1: \sigma_1^2 < \sigma_2^2$$

$$(iii) H_0: \sigma_1^2 = \sigma_2^2 \text{ vs. } H_1: \sigma_1^2 \neq \sigma_2^2$$

$$\text{Test statistic, } F = \frac{s_1^2}{s_2^2} = \frac{\text{Larger sample variance}}{\text{Smaller sample variance}}$$

The larger of the two sample variances is placed in the numerator of test statistic, forcing the ratio to be at least 1.00.

Decision:

Case	Hypothesis	F Formula	Decision Rule
Upper	$H_1: \sigma_1^2 > \sigma_2^2$	$F_{cal} = \frac{S_1^2}{S_2^2}$	Reject if $F_{cal} \geq F_{\alpha, (n_1-1), (n_2-1)}$
Lower	$H_1: \sigma_1^2 < \sigma_2^2$	$F_{cal} = \frac{S_2^2}{S_1^2},$ as $S_2^2 > S_1^2$	Reject if $F_{cal} \geq F_{\alpha, (n_2-1), (n_1-1)}$
Two-tailed	$H_1: \sigma_1^2 \neq \sigma_2^2$	$F_{cal} = \frac{S_1^2}{S_2^2}$	Reject if $F_{cal} \geq F_{\frac{\alpha}{2}, (n_1-1), (n_2-1)}$

Problem 1. Lammers Limos offers limousine service from Government Center in downtown Toledo, Ohio, to Metro Airport in Detroit. The president of the company is considering two routes. One is via U.S. 25 and the other via I-75. He wants to study the time it takes to get to the airport using each route and compare the results. He collected the following sample data. Using the .10 significance level, is there a difference in the variation in the driving times for the two routes?

U.S. Route 25	Interstate 75
52	59
67	60
56	61
45	51
70	56
54	63
64	57
	65

The null and alternate hypothesis

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

U.S. ROUTE 25

$$\bar{x} = \frac{\Sigma x}{n} = \frac{408}{7} = 58.29 \quad s = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}} = \sqrt{\frac{485.43}{7 - 1}} = 8.9947$$

INTERSTATE 75

$$\bar{x} = \frac{\Sigma x}{n} = \frac{472}{8} = 59.00 \quad s = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}} = \sqrt{\frac{134}{8 - 1}} = 4.3753$$

Test statistic,

$$F_{cal} = \frac{S_1^2}{S_2^2} = \frac{(8.9947)^2}{(4.3753)^2} = 4.23$$

Let, level of significance, $\alpha = 0.10$. As this is a two-tailed test, so we divide the level of significance (.10) by 2 and get $\frac{\alpha}{2} = \frac{0.10}{2} = 0.05$. Using the F table, we find the critical value using the degrees of freedom in the numerator, $n_1 - 1 = 7 - 1 = 6$ df, and the degrees of freedom in the denominator $n_2 - 1 = 8 - 1 = 7$ df as follows:

$$F_{critical} = F_{\frac{\alpha}{2}, (n_1 - 1), (n_2 - 1)} = F_{0.05, 6, 7} = 3.87.$$

Decision: As $F_{cal} > F_{critical}$, we may reject H_0 i.e., we may decide to reject the null hypothesis as the test statistic (4.23) exceeds the critical value (3.87). Therefore, we conclude there is a difference in the variation in the time to travel the two routes.

Exercise 1: Two machines are used to fill chemical bottles. The weights (in grams) of bottles filled by each machine are recorded.

Machine A ($n_1 = 8$): 52, 55, 50, 54, 53, 56, 51, 57

Machine B ($n_2 = 10$): 48, 49, 47, 50, 46, 51, 45, 49, 48, 47

Test at the 5% level of significance whether the variances are equal.

Exercise 2: Two laboratories measure the same chemical concentration. You want to check whether Lab A is less consistent (more variance) than B.

Lab A ($n_1 = 7$): 102, 110, 98, 115, 95, 108, 112

Lab B ($n_2 = 9$): 100, 101, 99, 98, 102, 100, 101, 99, 100

Test at $\alpha = 0.05$.

Exercise 3: Two processes are used to produce fiber. The manufacturer believes Process A is more consistent.

Process A ($n_1 = 6$): 25, 26, 24, 25, 26, 25

Process B ($n_2 = 8$): 23, 28, 22, 29, 24, 30, 21, 27

Test at $\alpha = 0.05$:

Comparison of k ($k > 2$) Independent Sample Means

If we want to test three or more sample means, it is a case of Analysis of Variance (ANOVA), which is a very powerful method of statistical analysis. We use F-test in such situations.

Suppose that, we have k independent samples from k populations, where the sample from the i^{th} population is of size n_i and μ_i is the mean of the i^{th} population.

We want to test the hypothesis-

$H_0: \mu_1 = \mu_2 = \dots = \mu_k$ against

H_1 : At least two means are unequal

Assumption:

K random samples have been drawn independently from k normal populations with the same variance σ^2 .

The sample observations may be denoted as follows:

	1 st Sample	2 nd Sample			K th Sample
	x_{11}	x_{21}			x_{k1}
	x_{12}	x_{22}			x_{k2}
	:	:	:		:
	:	:	:		:
	x_{1n_1}	x_{2n_1}			x_{kn_1}
Total	T_1	T_2			T_k
Mean	$\bar{x}_1$	$\bar{x}_2$			$\bar{x}_k$

The test statistic is,

$$F = \frac{S_B/k-1}{S_W/N-k} \sim F_{(k-1),(N-k)} \text{ with } (k-1) \text{ and } (N-k) \text{ degrees of freedom.}$$

Here, to calculate S_B and S_W , let us define-

$$\text{Grand total, } G = \sum_i \sum_j x_{ij} = \sum_i T_i$$

$$N = \sum_i n_i$$

$$\text{Correction term, } CT = \frac{G^2}{N}$$

$$\text{Total variation, } S_T = \sum_i \sum_j x_{ij}^2 - CT$$

$$\text{Between variation, } S_B = \sum_{i=1}^k \frac{T_i^2}{n_i} - CT$$

$$\text{Within variation, } S_W = S_T - S_B$$

Let, the level of significance is α .

The critical value of F is $F_{\alpha, (k-1), (N-k)}$.

Comment: If $F > F_{\alpha, (k-1), (N-k)}$, we may reject the null hypothesis.

Example 1 A chemical engineer wants to compare the mean yield (%) from three different reactor types.

Reactor A	Reactor B	Reactor C
72	65	78
75	68	82
78	70	85
74	72	88
76	69	90

Test whether the mean yields among the reactors are different. Also make pairwise comparisons.

Solution:

Hypotheses

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

$$H_1 : \text{At least one mean is different}$$

Totals and Means

$$T_1 = 72 + 75 + 78 + 74 + 76 = 375, \quad \bar{x}_1 = 75$$

$$T_2 = 65 + 68 + 70 + 72 + 69 = 344, \quad \bar{x}_2 = 68.8$$

$$T_3 = 78 + 82 + 85 + 88 + 90 = 423, \quad \bar{x}_3 = 84.6$$

$$G = T_1 + T_2 + T_3 = 375 + 344 + 423 = 1142$$

$$N = n_1 + n_2 + n_3 = 15$$

Correction Term (CT)

$$CT = \frac{G^2}{N} = \frac{1142^2}{15} = \frac{1304164}{15} = 86944.27$$

Between-Sample Variation (SB)

$$\begin{aligned} SB &= \sum_{i=1}^k \frac{T_i^2}{n_i} - CT \\ &= \frac{375^2}{5} + \frac{344^2}{5} + \frac{423^2}{5} - 86944.27 \\ &= 28125 + 23667.2 + 35772.6 - 86944.27 \\ SB &= 633.53 \end{aligned}$$

Total Variation (ST)

$$\begin{aligned} ST &= \sum_i \sum_j x_{ij}^2 - CT \\ \sum x_{ij}^2 &= 72^2 + 75^2 + \cdots + 90^2 = 87713 \\ ST &= 87713 - 86944.27 = 768.73 \end{aligned}$$

Within-Sample Variation (SW)

$$SW = ST - SB = 768.73 - 633.53 = 135.20$$

F-Statistic

$$F = \frac{S_B/(k-1)}{S_W/(N-k)} = \frac{633.53/2}{135.20/12} = \frac{316.77}{11.27} = 28.10$$

◆ ANOVA Table

Source	SS	df	MS	F
Between	633.53	2	316.77	28.10
Within	135.20	12	11.27	
Total	768.73	14		

Decision

Critical value at $\alpha = 0.05$:

$$F_{0.05,(2,12)} \approx 3.89$$

Since

$$F_{calc} = 28.10 > 3.89$$

Reject H_0

The $k = 3$ population means are **not equal**

There is a **significant difference** among the mean yields of the three reactors. At least one reactor performs differently.

For pairwise comparison

Hypothesis: $H_0: \mu_i = \mu_j$ vs. $H_1: \mu_i \neq \mu_j; i \neq j = 1, 2, \dots, k$.

Test statistic, $t = \frac{|\bar{x}_i - \bar{x}_j|}{\sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}}$, where $MSE = S_w / (N - k)$

Critical value at $\alpha, t_{\frac{\alpha}{2}, (N-k)}$. At $\alpha = 0.05$, the critical value is $t_{0.025, 12} = 2.179$.

Decision: If $|t_{cal}| \geq |t_{critical}| \Rightarrow \text{Reject } H_0$,

Otherwise no enough evidence to reject H_0 .

Comparison	Mean Diff	t	$t_{0.025, 12}$	Comment
A vs B	$75 - 68.8 = 6.2$	2.91	2.179	Reject H_0

A vs C	$75 - 84.6 = -9.6$	4.51	2.179	<i>Reject H_0</i>
B vs C	$68.8 - 84.6 = -15.8$	7.42	2.179	<i>Reject H_0</i>

Exercises.

1. A chemical plant tests three catalysts for % yield.

Catalyst A	Catalyst B	Catalyst C
68	72	75
70	74	78
73	76	80
75	78	82
77		85
		88

Tasks:

1. Compute sample means.
2. Perform one-way ANOVA at $\alpha=0.05$.
3. Construct the ANOVA table.
4. Test whether all means are equal.
5. Make pairwise comparisons.
6. Interpret the results.

2. Energy use (kWh) of three processes:

Process 1	Process 2	Process 3
420	450	390

Process 1	Process 2	Process 3
430	460	400
440	470	410
445		420
455		430
		440

Tasks:

1. Find group means and grand mean.
2. Prepare the ANOVA table.
3. Test at $\alpha=0.01$.
4. Make pairwise comparisons.
5. Which process is most energy-efficient?